Notes: Schmidt, Timmermann, and Wermers (AER, 2016)

Runs on Money Market Mutual Funds

Contents

1 Big picture

- **Question.** Why did some money market mutual funds experience run-like withdrawals after Lehman Brothers failed, while other seemingly similar funds did not? The paper asks whether the run was driven only by weak portfolio fundamentals or also by strategic complementarities among investors.
- **Setting.** The empirical setting is September 2008. The Reserve Primary Fund broke the buck after Lehman failed, and run-like behavior spread especially to prime institutional money market mutual funds. Prime funds hold private short-term debt such as commercial paper, while government funds hold Treasury and agency securities.
- **Core data contribution.** The authors use daily iMoneyNet data at the individual share-class level. This is much finer than fund-level monthly data. The share-class structure is crucial because different investor types can hold different share classes in the same fund while sharing the same underlying portfolio.
- **Key identification idea.** Within a single fund, low-expense-ratio institutional share classes and high-expense-ratio institutional share classes hold pro rata claims on exactly the same assets. Low expense ratios require larger investment minimums and proxy for more sophisticated, more attentive, larger-scale investors.
- **Theoretical mechanism.** The paper builds a simple coordination model with informed and uninformed investors. A run becomes more likely when more investors are informed and strategic, because investors have a stronger incentive to redeem before others impair the fund's liquidity.
- **Main empirical result.** Sophisticated investors run more. Outflows are larger from low-expense-ratio institutional share classes, larger in funds with a higher fraction of sophisticated investors, and especially large when sophisticated investors are surrounded by other sophisticated investors.
- **Dynamic result.** Sophisticated investors react to the past and contemporaneous behavior of less sophisticated investors when those less sophisticated investors represent a large share of the fund. This is consistent with strategic interaction, not just individual reactions to fundamentals.
- **Fixed-NAV result.** Comparing money market funds with ultra-short bond funds suggests that the fixed $1 NAV structure amplified run incentives. Ultra-short funds with floating NAVs had negative returns but less severe immediate outflows.
- **Distributional result.** The run was concentrated in the left tail. Quantile regressions show that investor composition, fund size, and flow volatility matter most for the worst outflow realizations.
- **Economic takeaway.** Runs on cash-like pooled vehicles are not only about asset risk. They also depend on who the other investors are, how attentive they are, and whether the payoff structure gives investors a first-mover advantage.

2 Incremental contribution of the paper

- **From aggregate MMMF outflows to micro run mechanics.** Earlier work documents that money market funds were fragile during the crisis. This paper shows how run incentives operate at the share-class level and across different investor clienteles inside the same fund.

- **A direct empirical test of strategic complementarities.** The paper maps predictions from a coordination/global-games model to observable cross-sectional patterns in investor flows. It tests not just whether risky funds lose assets, but whether investor composition changes run incentives.
- **Within-fund identification using multiple share classes.** The same-fund design holds constant the portfolio, adviser, sponsor support, and many unobserved fund-level fundamentals. This is the central empirical advantage of the paper.
- **Investor sophistication measurement.** The paper uses expense ratios and minimum investment requirements to classify institutional investors by likely sophistication and attentiveness. Low expense ratios proxy for large, sophisticated investors with more skin in the game.
- **Dynamic evidence at the daily frequency.** The paper studies how flows from sophisticated and unsophisticated investors interact day by day. This directly speaks to the speed and feedback nature of runs.
- **Fixed NAV versus floating NAV evidence.** By comparing prime MMMFs with same-complex ultra-short bond funds, the paper gives suggestive evidence that fixed NAV accounting strengthens first-mover incentives.
- **Tail-risk measurement.** The dynamic quantile regressions quantify how fund characteristics affect the distribution of outflows, especially the extreme left tail where run risk is most important.
- **Policy contribution.** The paper warns that reforms can have offsetting effects. Floating NAVs may weaken first-mover incentives, but separating institutional and retail investors may make institutional funds more run-prone by concentrating sophisticated investors together.

3 Data and measurement

3.1 Institutional background

- Money market mutual funds are regulated mutual funds that historically maintained a stable \$1 share price using amortized-cost accounting under Rule 2a-7.
- They offer cash-like liabilities but hold assets that can become illiquid under stress. The liquidity mismatch is smaller than in banks, but the fixed NAV gives investors a reason to redeem early if they fear losses or forced sales.
- The key crisis event is the Lehman bankruptcy and the Reserve Primary Fund's breaking of the buck. This generated a sudden public signal about risk in prime money funds.
- Prime institutional funds experienced the most severe outflows; prime retail funds were much more stable; government funds received inflows as investors fled to safety.

Interpretation: The paper treats the Lehman episode as a natural experiment: a large, unexpected shock made investors suddenly care about both portfolio risk and the behavior of other investors.

3.2 iMoneyNet share-class data

- The main data come from iMoneyNet and cover daily information on more than 2,000 US registered MMMFs from February to December 2008.
- The authors focus mainly on prime money funds because these funds hold commercial paper and other non-government short-term instruments.
- The data include share-class type, daily total net assets, fund family/adviser identity, investment objective, expense ratio, average maturity, and weekly portfolio sector breakdowns.
- iMoneyNet covers about 93.5% of the dollar value of the US prime MMMF universe at the end of 2008.
- The Reserve complex is excluded because the Reserve Primary Fund is the trigger event and other funds in the same complex may have brand-contagion effects.

Interpretation: Daily share-class data make it possible to observe who runs, when they run, and whether their actions depend on other investor types in the same fund.

3.3 Investor sophistication proxy

- The paper uses the share-class expense ratio as a negative proxy for investor sophistication.
- The logic is institutional: lower expense ratios are generally available only to investors who meet larger minimum investment requirements.
- Table 1 confirms this empirically. A higher expense ratio is strongly associated with a lower minimum required investment.
- The benchmark cutoff often used in the paper is 35 basis points per year. Institutional share classes with expense ratios at or below this cutoff are classified as sophisticated or large-scale.

Interpretation: The paper does not directly observe investor information sets. Instead, it uses the fund's price menu: lower-cost institutional classes screen in larger investors who are more likely to monitor risks and respond quickly.

3.4 Flow measurement and control variables

- The main dependent variable is share-class flow, measured as the change in logged assets under management. In the paper's tables, flows are often multiplied by 100.
- Positive flow means assets increase; negative flow means outflows. Some figures report net outflows, so signs are reversed for visual interpretation.
- Main fund-level controls include log fund size, average gross yield, liquid asset share, and fund business outside prime institutional money funds.
- Average gross yield proxies for portfolio risk-taking: higher yields may indicate riskier underlying assets.
- Liquid asset share captures asset-side liquidity, while %Soph captures liability-side funding fragility.

Interpretation: The controls are designed to separate investor-composition effects from differences in fund fundamentals. The paper's main claim is not that fundamentals are irrelevant, but that investor composition has independent predictive power.

3.5 Ultra-short bond fund comparison data

- To study the role of fixed NAV, the paper compares MMMFs with ultra-short bond funds.
- Ultra-short bond funds invest in similar short-term fixed-income instruments but have floating NAVs.
- The ultra-short analysis uses CRSP mutual fund data because CRSP allows the authors to link funds within the same complex.
- Since CRSP asset data are monthly, this comparison is based on September 2008 monthly flows rather than daily flows.

Interpretation: The ultra-short comparison is not a perfect experiment because the investor base and assets differ. It is best read as suggestive evidence about whether the fixed $1 NAV amplifies first-mover incentives.

4 Models and empirical specifications

4.1 Simple coordination model

The model is a static coordination game with regime change. A fund maintains the status quo, interpreted as the stable $1 NAV, unless enough investors withdraw.

Let $a_i = 1$ if investor i withdraws and $a_i = 0$ otherwise. Let A be the aggregate mass of withdrawals, and let θ denote fundamentals, where higher θ means stronger fundamentals. The payoff is

$$U(a_i, A, \theta) = a_i \left(\mathbf{1}\{A > \theta\} - c \right).$$

- If the fund breaks the status quo and the investor withdrew, the payoff is $1 - c$.
- If the investor withdraws but the status quo survives, the investor pays cost c.
- The strategic complementarity is that withdrawing is more attractive when more other investors withdraw.

A fraction μ of investors are informed. Informed investors receive noisy private and public signals about θ and follow a threshold rule: withdraw if their private signal is below $x^*(z)$. The aggregate attack is

$$A(\theta, z) = \mu \Phi \left(\sqrt{\alpha_x}(x^*(z) - \theta) \right).$$

Interpretation: The key comparative static is that increasing the mass of informed or sophisticated investors makes coordination on the no-run outcome harder to sustain. More attentive investors can make the fund more fragile because they react strategically to each other.

4.2 Testable predictions

The paper translates the model into three empirical predictions using two investor types: sophisticated investors S and unsophisticated investors U.

1. **Within-fund sophistication prediction.** Within the same fund, outflows from sophisticated investors should be larger than outflows from unsophisticated investors after the shock.
2. **Investor-composition prediction.** Holding investor type fixed, outflows should be larger in funds with a higher fraction of sophisticated investors.
3. **Interaction prediction.** The difference between sophisticated and unsophisticated outflows should be larger when the fund contains a higher fraction of sophisticated investors.

Interpretation: These predictions are powerful because two of them can be tested within funds, where the underlying portfolio and sponsor are held fixed.

4.3 Cross-sectional share-class regressions

The first empirical specification tests whether lower expense ratio share classes have larger outflows:

$$Flow_{ij} = \alpha + \beta ER_{ij} + \delta_j + \varepsilon_{ij},$$

where $Flow_{ij}$ is the crisis-week flow for share class i in fund j, ER_{ij} is the expense ratio, and δ_j can be a fund fixed effect.

The second specification tests whether the fraction of sophisticated investors predicts outflows:

$$Flow_{ij} = \alpha + \beta \% Soph_j + \gamma ER_{ij} + X_j' \Lambda + \varepsilon_{ij}.$$

The interaction version tests the third prediction:

$$Flow_{ij} = \alpha + \beta \% Soph_j + \gamma ER_{ij} + \eta(\% Soph_j \times ER_{ij}) + X_j' \Lambda + \varepsilon_{ij}.$$

- A positive coefficient on ER in flow regressions means high-expense-ratio share classes have less negative flows, i.e., lower outflows.
- A negative coefficient on $\% Soph$ means funds with more sophisticated investors experience larger outflows.
- A positive interaction between $\% Soph$ and ER means the sophistication gap becomes larger in more sophisticated funds.

4.4 Fixed-NAV comparison with ultra-short bond funds

For the fixed-NAV analysis, the paper estimates regressions of ultra-short bond fund flows on same-complex prime MMMF flows:

$$Flow_{ic}^{UltraShort} = \alpha + \beta Flow_c^{PrimeMMMF} + X_{ic}' \Gamma + \varepsilon_{ic}.$$

- The comparison is within fund complexes, which partially controls for sponsor reputation and management style.

- If fixed NAVs do not matter, and ultra-short assets are at least as impaired, ultra-short funds should have outflows at least as severe as prime MMMFs.
- The paper finds that ultra-short outflows are more muted, consistent with fixed NAVs increasing run incentives in MMMFs.

4.5 Dynamic interactions between investor types

The daily dynamic specification separates low-expense-ratio institutional share classes from high-expense-ratio institutional and retail share classes within the same fund. The dependent variable is flow from sophisticated share classes:

$$Low_{it} = \alpha + \rho Low_{i,t-1} + \phi High_{i,t-1} + \psi(High_{i,t-1} \times \%High_{i,t-2}) + X'_{it}\Gamma + \tau_t + \varepsilon_{it}.$$

The important term is $High_{i,t-1} \times \%High_{i,t-2}$. It asks whether sophisticated investors react more strongly to unsophisticated outflows when unsophisticated investors are a large part of the fund.

4.6 Dynamic quantile regressions

The paper also estimates dynamic panel quantile regressions for daily fund-level flows. The goal is to understand how observable fund characteristics affect not only the average outflow but also the extreme left tail of the flow distribution.

- The key variables are %Soph, flow volatility, average gross yield, and fund size.
- The simulation exercise reports how one-standard-deviation perturbations shift cumulative flow quantiles over Lehman week.
- This is important because run risk is a tail event: the most important policy question is what happens to the worst-outflow funds, not just to the median fund.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies strong empirical patterns consistent with strategic complementarities among investors in money market funds.
- The strongest design compares different share classes inside the same fund. This holds constant portfolio holdings, fund adviser, sponsor support, and many unobserved fund-level characteristics.
- The evidence is most convincing for the claim that investor composition matters for run risk over and above observable asset-side fundamentals.

Interpretation: The design does not literally randomize investor composition. It is an observational design built around a sudden crisis shock and unusually rich within-fund variation.

5.2 The Lehman shock as a natural experiment

- The Lehman bankruptcy and Reserve Primary Fund event created a public shock to beliefs about prime MMMF safety.
- Because the shock was sudden, it is useful for studying fast strategic behavior.
- The cross-sectional question is why some funds suffered severe outflows while others did not.

Interpretation: The paper's empirical leverage comes from the fact that the same aggregate shock hit funds with different investor mixes and portfolio characteristics.

5.3 Expense ratios and investor sophistication

- Expense ratios are not themselves information; they are a sorting device.
- Low expense ratios are associated with high minimum investment requirements, larger accounts, and stronger incentives to monitor.
- The paper validates this relationship directly in Table 1 before using expense ratios throughout the main tests.

Interpretation: This measurement step is essential. Without it, differences in share-class behavior could be interpreted simply as fee effects rather than investor-clientele effects.

5.4 Strategic complementarities versus fundamentals

- A central alternative explanation is that sophisticated investors simply know more about bad assets in certain funds.
- The paper addresses this by using fund fixed effects in within-fund regressions and by controlling for yield, liquidity, size, and sponsor business exposure.
- The dynamic results are especially useful because sophisticated investors respond to less sophisticated investors' flows in a way that depends on the latter group's share of the fund.

Interpretation: Fundamentals matter, but the paper argues that fundamentals alone cannot explain why investor flows vary systematically with the composition and actions of other investors in the same fund.

5.5 Fixed NAV and external validity

- The ultra-short bond fund comparison helps isolate the role of the fixed $1 NAV.
- The evidence is suggestive rather than definitive because ultra-short funds have different assets and clienteles.
- Still, the comparison aligns with the idea that a fixed NAV creates a stronger first-mover advantage than a floating NAV.

Interpretation: The fixed-NAV analysis complements the main share-class tests. The main tests identify investor-composition effects; the ultra-short analysis points to the payoff structure that amplifies those effects.

5.6 Limitations of the empirical design

- Investor sophistication is proxied rather than directly observed.
- Some unobserved portfolio risk could remain even after controls, especially because asset quality deteriorated quickly during the crisis.
- The ultra-short comparison is not a randomized fixed-versus-floating NAV experiment.
- The results come from an extreme crisis episode, so magnitudes may not generalize to normal times.

Interpretation: The paper is best read as a carefully structured observational study of run mechanisms during a crisis, not as a fully randomized causal experiment.

6 Tables and figures

6.1 Figure 1: Money market mutual fund flows in September–October 2008

Read:

- Panel A plots daily flows by category: prime institutional, prime retail, government institutional, and government retail.
- Prime institutional funds experience the sharpest one-day outflow, exceeding 10% of assets on September 17.
- Panel B shows cumulative dollar flows. Prime institutional outflows reach roughly $400 billion over the first two weeks of the crisis.
- Government funds receive inflows, consistent with a flight to safety.
- Panel C shows that the run is highly uneven across prime institutional funds. Some funds lose more than 50% of assets, while others have modest outflows or even inflows.

Economic message: Figure 1 motivates the whole paper. The aggregate run is large, but the cross-sectional dispersion is the key fact: investors did not flee all prime funds equally. The rest of the paper explains this dispersion using fund fundamentals and investor composition.

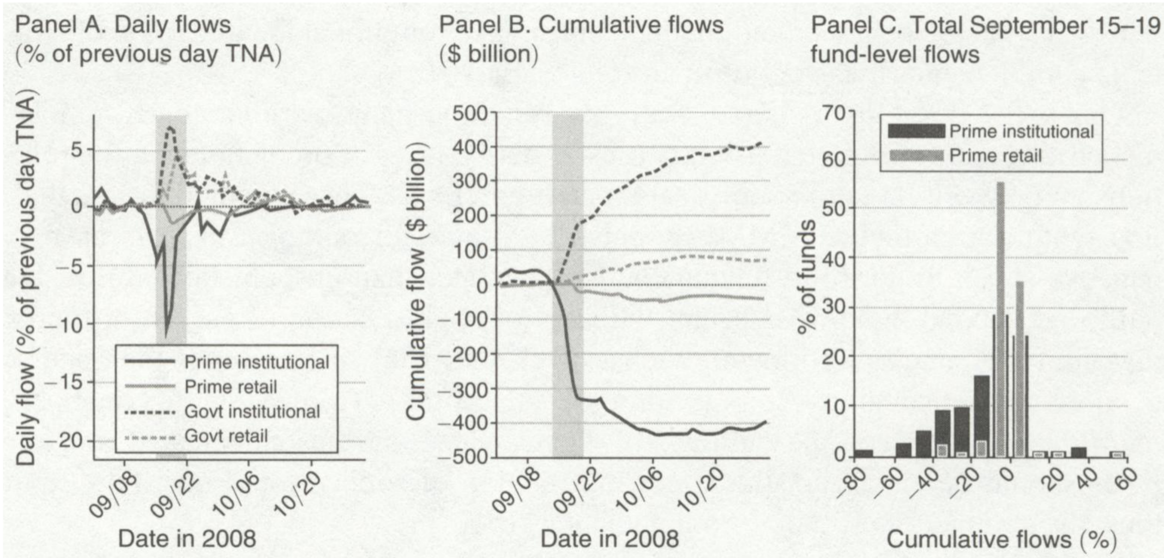

Figure 1. Money Market Mutual Fund Flows in September–October 2008

Notes: This figure plots summary statistics for fraction flows (one-day difference in log assets) to/from money market mutual fund share classes in different aggregate categories during September and October 2008. For each category, panel A displays the daily percentage flow, while panel B plots the cumulative flow (in billions of dollars) over this period. Panel C presents a histogram of the cross-sectional distribution of fund-level percentage change in total assets under management for share classes within each category during September 15–19 (i.e., flows during the MMMF crisis week following the failure of Lehman Brothers, which is indicated by shading).

Figure 1: Money market mutual fund flows in September–October 2008.

6.2 Table 1 and Figure 2: Measuring sophistication and summarizing theoretical tests

Read:

- Table 1 validates the expense-ratio proxy for investor sophistication. Lower expense ratios are strongly associated with higher minimum investment levels.
- For prime funds, a one percentage point increase in the expense ratio is associated with a large decline in the minimum investment requirement.
- The discrete specification shows that low-expense-ratio share classes have much larger minimums than high-expense-ratio share classes.
- Figure 2 converts the theory into a simple visual test. Large institutional low-ER share classes have the largest outflows, especially when they are in funds with a high fraction of other large institutional investors.
- The pattern in Figure 2 lines up with all three predictions: sophisticated investors run more; funds with more sophisticated investors have larger outflows; and the sophistication gap is larger in sophisticated-investor-heavy funds.

Economic message: Table 1 builds the measurement foundation. Figure 2 then shows why the measurement matters: the investors most likely to be informed and strategic are precisely the investors who redeem most aggressively during the run.

TABLE 1—REGRESSIONS OF MINIMUM INVESTMENT LEVELS ON SHARE CLASS EXPENSE RATIOS

	Prime only		Government only		Prime and government	
Panel A. Continuous expense ratio measure						
Share class expense ratio (pp/yr)	−6.05	−5.23	−7.64	−4.91	−6.80	−5.04
	(−2.668)	(−3.052)	(−4.683)	(−3.934)	(−4.975)	(−4.977)
R^2	0.055	0.907	0.102	0.876	0.075	0.892
Fund dummies	No	Yes	No	Yes	No	Yes
Observations	226	226	265	265	491	491
Degrees of freedom	224	132	263	175	489	308
Panel B. Discrete expense ratio measure						
Dummy: Expense ratio < 35 bp/yr	2.68	2.06	3.01	1.64	2.83	1.81
	(3.290)	(3.293)	(5.736)	(3.955)	(6.071)	(5.142)
R^2	0.055	0.905	0.086	0.868	0.069	0.887
Fund dummies	No	Yes	No	Yes	No	Yes
Observations	226	226	265	265	491	491
Degrees of freedom	224	132	263	175	489	308

Notes: This table presents evidence that lower expense ratios are associated with higher investment minimums in the cross section of institutional share classes of MMMFs with available data on minimums, measured as of September 12, 2008. In panel A, each cell in the table provides the coefficient and t-statistic from a univariate regression of the log of the minimum dollar investment level of a given share class on its expense ratio, measured in pp/yr). Each pair of columns corresponds to different subsets of the institutional share class population: prime institutional share classes, government institutional share classes, and both populations pooled, with and without fund fixed effects, respectively. In panel B, we replace the continuous expense ratio with a dummy variable which equals one when the expense ratio is less than 35 bp/yr, and zero, otherwise.

(a) Table 1: Expense ratios and minimum investment levels.

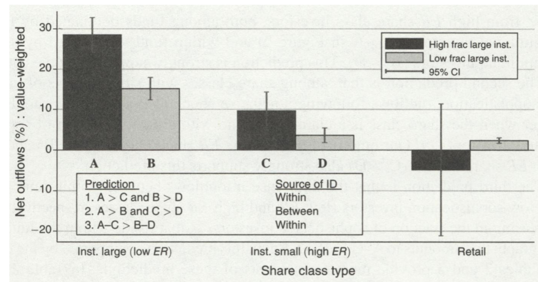

FIGURE 2. SUMMARY OF TESTS OF THEORETICAL PREDICTIONS

Notes: This graph shows net outflows (the percentage change in total fund assets under management multiplied by −1) by share class type, across all prime MMMFs from September 15 to September 19. First, each share class within each prime fund is placed in one of three bins: (i) retail share classes, and institutional share classes having an expense ratio (ii) above or (iii) no more than 35 bps/yr. Next, each of these three bins is subdivided, according to whether or not the share class is a claim on a prime fund with a high fraction—at least 75 percent—of its investment dollars (across all share classes, retail and institutional) represented by institutional share classes with an expense ratio no more than 35 bps/yr. Value-weighted average net outflows with 95 percent confidence intervals (CI), are shown for each of the six bins.

(b) Figure 2: Summary of theoretical predictions.

6.3 Tables 2 and 3: Cross-sectional share-class evidence

Read:

- Table 2 tests prediction (i). Share-class expense ratio is positively related to flows, meaning higher-ER classes have less negative flows and lower-ER classes have larger outflows.
- The relation survives complex and fund fixed effects, so it is not only a comparison across different portfolios or sponsors.
- The nonparametric bins show that the lowest-ER share classes have especially large crisis-week outflows.
- Table 3 tests prediction (ii) and prediction (iii). Outflows from sophisticated share classes are larger in funds with a higher fraction of sophisticated investors.
- The coefficient on %Soph is negative across alternative expense-ratio cutoffs, and the absolute magnitude grows as the cutoff becomes more restrictive.
- The interaction between %Soph and expense ratio is positive and significant, indicating that the gap between low-ER and high-ER investors is larger in funds with more sophisticated investors.

Economic message: These tables are the core cross-sectional evidence. They show that investor behavior depends not only on a share class's own sophistication, but also on the sophistication of the fund's other investors.

Table 2—Regressions of Prime Institutional Share Class–Level Flows on Expense Ratio

	(1)	(2)	(3)	(4)	(5)	(6)
Share class expense ratio	10.89	11.14	12.42			
	(5.181)	(4.153)	(3.153)			
Indicators:						
15 < ER ≤ 35				−15.59	−21.29	−20.85
				(−4.865)	(−5.122)	(−2.682)
ER ≤ 15				−34.88	−36.09	−38.97
				(−3.913)	(−2.894)	(−2.395)
Dummies	None	Complex	Fund	None	Complex	Fund
Clustering	Fund	Complex	Fund	Fund	Complex	Fund
Observations	258	258	258	258	258	258
Degrees of freedom	256	191	135	255	190	134
R^2	0.057	0.369	0.565	0.065	0.381	0.575

Notes: To test model prediction (i), this table presents estimated coefficients from OLS regressions of the change in logged prime institutional share class assets under management (i.e., flows as a fraction of lagged assets under management, × 100) from September 15 to September 19 on that share class's expense ratio (*ER*), which has been normalized by its cross-sectional standard deviation. Depending on the specification, we include no fixed effects, complex fixed effects, or fund fixed effects. The final three columns replace the continuous share class *ER* with two dummy variables for different ranges of *ER*; the omitted category includes all prime institutional share classes with an *ER* > 35 bp/yr (*t*-statistics in parentheses).

(a) Table 2: Flows and share-class expense ratios.

Table 3—Regressions of Prime Institutional Share Class–Level Flows on Fund and Investor Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
%Soph (25 bp/yr)	−42.2							
	(−2.47)							
%Soph (30 bp/yr)		−36.4						
		(−2.80)						
%Soph (35 bp/yr)			−33.8			−12.6	−50.3	
			(−3.01)			(−1.04)	(−3.77)	
%Soph (50 bp/yr)				−31.2				
				(−3.12)				
%Soph (150 bp/yr)					−28.6			
					(−2.92)			
%Soph (35 bp/yr) × ER							17.8	31.5
							(3.97)	(3.34)
Share class expense ratio	8.4	16.8	16.0	15.0	10.4	5.7	−2.7	−10.0
	(0.54)	(1.56)	(1.99)	(4.02)	(4.93)	(1.28)	(−1.19)	(−1.84)
log total fund assets	−12.7	−10.8	−10.3	−8.2	−7.5	−1.6	−7.5	
	(−2.93)	(−2.80)	(−3.25)	(−3.22)	(−3.36)	(−0.67)	(−3.48)	
Average gross yield	−1.1	−2.1	−2.5	−5.0	−3.7	−6.9	−3.4	
	(−0.15)	(−0.33)	(−0.41)	(−0.78)	(−0.71)	(−1.26)	(-0.64)	
Liquid asset share	4.9	2.8	2.8	0.1	1.4	−0.9	1.0	
	(0.73)	(0.48)	(0.52)	(0.01)	(0.30)	(−0.18)	(0.23)	
Fund business	−0.5	−1.0	−1.2	−1.6	−1.7	−2.0	−1.3	
	(−0.11)	(−0.25)	(−0.35)	(−0.56)	(−0.64)	(−0.90)	(−0.53)	
Sample selection	Sophisticated institutional: ER ≤ cutoff					ER > 35 bp/yr	All	All
Fixed effects	None	None	None	None	None	None	None	Fund
Observations	118	138	161	215	258	97	258	258
Degrees of freedom	111	131	154	208	251	90	250	134
R^2	0.080	0.089	0.094	0.111	0.118	0.076	0.128	0.582

Notes: To test model prediction (ii), this table presents estimated coefficients from OLS regressions of the change in logged share class–level assets under management (i.e., flows as a fraction of lagged assets under management (× 100)) for each prime institutional money market share class from September 15 through September 19 on expense ratio (*ER*); percent large-scale institutions (*%Soph*; defined as the fraction of total fund investment dollars owned by investors of institutional share classes with *ER* ≤ a particular cutoff: 25 bp for the first column); logged fund-level size; average annualized 7-day SEC gross yield over the six month period prior to the crisis (March–August 2008); liquid asset share, estimated daily by computing dollar proportion Treasury and US agency securities plus repo investments, plus (estimated) maturing securities, minus net redemptions; and fund business, defined as one minus proportion (by value) of fund complex aggregate mutual fund assets that are represented by prime institutional share classes. The bottom five variables in the table have each been divided by their most recently available cross-sectional standard deviations to normalize. Different columns correspond with different cutoffs and different sample selection criteria. To test prediction (iii) of our theoretical model, the final two columns add an interaction term between share class expense ratio and *%Soph* at the fund level. Note that the specification in the last column includes fund fixed effects. *t*-statistics are indicated in parentheses.

(b) Table 3: Flows, fund characteristics, and investor composition.

6.4 Table 4: Cumulative flows during Lehman week

Read:

- Table 4 repeats the cross-sectional tests using cumulative flows over progressively longer parts of Lehman week.
- The differences are already visible on September 15 and become larger as the week progresses.
- The expense-ratio effect grows from the early crisis to the peak crisis.
- The %Soph and interaction effects also strengthen over the week, consistent with a run that builds dynamically as investors observe stress and react.

Economic message: Table 4 shows that the run was not simply a single one-day mechanical reaction. Sophisticated-investor effects accumulated as the crisis unfolded, which is consistent with dynamic strategic behavior.

Table 4—Regressions of Share Class–Level Flows on Fund and Investor Characteristics: Cumulative Flows over the Course of Lehman Week

	(1)	(2)	(3)	(4)	(1)	(2)	(3)	(4)
	Panel A. Prediction (i)				*Panel C. Predictions (i)–(iii)*			
	9/15	9/15–16	9/15–17	9/15–18	9/15	9/15–16	9/15–17	9/15–18
Share class expense ratio	1.6	3.0	7.8	10.1	−1.2	−2.7	−6.1	−8.4
	(2.35)	(2.26)	(3.01)	(3.26)	(−1.12)	(−1.40)	(−1.69)	(−1.97)
%Soph (35 bp/yr) × ER					4.1	8.0	19.5	26.1
					(2.21)	(2.48)	(3.48)	(3.94)
Specification	All institutional, fund F.E. (N = 258)				All institutional, fund F.E. (N = 258)			
R^2	0.542	0.538	0.436	0.521	0.552	0.555	0.461	0.546
	Panel B. Predictions (i)–(ii)				*Panel D. Predictions (i)–(iii)*			
	9/15	9/15–16	9/15–17	9/15–18	9/15	9/15–16	9/15–17	9/15–18
%Soph (35 bp/yr)	−6.4	−9.9	−19.1	−27.7	−8.8	−13.8	−29.5	−44.2
	(−2.63)	(−3.07)	(−3.92)	(−4.41)	(−2.93)	(−3.44)	(−4.77)	(−5.34)
%Soph (35 bp/yr) × ER					3.3	4.9	11.0	15.5
					(2.92)	(3.22)	(4.71)	(4.88)
Share class expense ratio	2.4	3.4	13.1	15.7	−0.9	−1.1	−2.2	−3.1
	(1.17)	(1.03)	(2.52)	(2.40)	(−1.59)	(−1.48)	(−1.84)	(−1.89)
log total fund assets	−1.5	−2.6	−5.7	−8.3	−1.0	−1.9	−4.0	−5.9
	(−2.77)	(−2.69)	(−3.47)	(−4.05)	(−2.75)	(−2.75)	(−3.51)	(−4.05)
Specification	ER ≤ 35 bp/yr, controls, no F.E. (N = 161)				All institutional, controls, no F.E. (N = 258)			
R^2	0.062	0.072	0.107	0.151	0.088	0.115	0.129	0.166

Notes: This table presents estimated coefficients (*t*-statistics) from OLS regressions of the change in the log of share class–level assets under management (i.e., flows as a fraction of lagged assets under management) for prime institutional money market funds (× 100), cumulated over parts of the period from September 15–18, 2008, on expense ratio, percent large-scale institutions (*%Soph*; calculated with 35 bp/yr cutoff), as well as the log of fund size. Regressions in panels B and D also control for average annualized gross yield, liquid asset share, and fund business; these (insignificant) coefficients are suppressed. See notes to Table 3 for more detailed variable descriptions. Expense ratio (*ER*) and size are divided by their cross-sectional standard deviations. Different columns denote different sample periods. Panels C and D add an interaction term between share class expense ratio and *%Soph* at the fund level.

Table 4: Cumulative share-class flows over Lehman week.

6.5 Table 5 and Figure 3: Fixed NAV and ultra-short bond funds

Read:

- Table 5 compares flows in ultra-short bond funds with same-complex prime MMMF flows during September 2008.
- Ultra-short bond funds have floating NAVs, while MMMFs at the time had stable \$1 NAVs.
- The coefficient on same-complex prime retail MMMF flow is positive but less than one, meaning ultra-short flows move in the same direction but with smaller amplitude.
- Figure 3 shows that ultra-short funds experienced sharply negative returns during Lehman week, but their outflows were less severe than the immediate MMMF run.

- The pattern is consistent with fixed NAVs amplifying run incentives in money funds.

Economic message: A floating NAV does not eliminate fragility, but it reduces the simple payoff discontinuity created by a stable \$1 price. The evidence suggests that the fixed NAV helped convert credit-quality concerns into immediate run incentives.

TABLE 5—REGRESSIONS OF SHARE CLASS–LEVEL FLOWS FOR SAME-COMPLEX ULTRA-SHORT BOND FUNDS ON FLOWS FOR PRIME MMMFS

	(1)	(2)	(3)	(4)
Complex prime retail MMF flow	0.51		0.58	
	(3.929)		(3.352)	
Complex prime MMF flow		0.33		0.18
		(2.112)		(1.795)
Institutional share class dummy	0.51	−1.06	0.74	0.46
	(0.139)	(−0.245)	(0.144)	(0.088)
Expense ratio	1.73	2.68	2.45	2.18
	(1.156)	(1.628)	(0.940)	(0.749)
Annualized yield	0.68	1.23	0.39	1.12
	(0.491)	(0.894)	(0.245)	(0.697)
log total fund assets	−5.99	−5.55	−7.08	−4.66
	(−3.387)	(−2.846)	(−2.660)	(−1.859)
Sample selection	All ultra-short funds		Ultra-short funds > \$10 MM	
Observations	45	50	36	39
R^2	0.285	0.226	0.264	0.129

Notes: This table presents estimated coefficients from OLS regressions of share class–level flows as a fraction of lagged assets under management (i.e., the change in log assets under management, adjusted for changes in market value × 100) for mixed (non-government) ultra-short bond fund institutional share classes during the month of September 2008 on flows over the same period from same-complex prime retail MMMF (models 1 and 3) or same-complex prime retail plus prime institutional MMMF share classes (models 2 and 4); in each case, these same-complex MMMF flows are aggregated to the complex level. In addition, each regression includes a dummy variable which equals one for an institutional share class, the share class expense ratio, the log of fund size (total net assets under management as of August 31, 2008) and its annualized gross yield (fund interest distributions over the previous 12 months, divided by current share price); the latter three variables have been divided by their cross-sectional standard deviations. All variables (including money market flows) are constructed using the CRSP mutual fund database.

(a) Table 5: Ultra-short bond fund flows and same-complex MMMF flows.

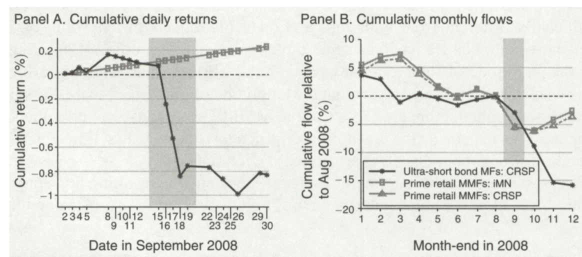

FIGURE 3. CUMULATIVE RETURNS AND FLOWS FOR MMMF AND ULTRA-SHORT BOND FUNDS

(b) Figure 3: Cumulative returns and flows for MMMFs and ultra-short bond funds.

6.6 Table 6: Dynamic interactions between investor types

Read:

- Table 6 studies daily flows from sophisticated low-ER share classes.
- The key explanatory variables are lagged flows from low-ER and high-ER investors, interacted with the fraction of the fund owned by high-ER or less sophisticated investors.
- The coefficient on $High_{i,t-1} \times \%High_{i,t-2}$ is positive and significant in the peak crisis and Lehman week specifications.
- This means sophisticated investors respond more strongly to less sophisticated investor outflows when less sophisticated investors represent a larger fraction of the fund.
- The contemporaneous interaction $High_{i,t} \times \%High_{i,t-1}$ is also large and significant, suggesting coordination can occur within the day.
- Liquid asset share matters more during the peak crisis, indicating that asset-side liquidity concerns became more important once the run intensified.

Economic message: Table 6 is important because it directly studies investor interaction. Sophisticated investors do not just react to their own signals; they react to the actions of other investor groups, and the strength of that reaction depends on how important those groups are within the fund.

Table 6—Determinants of Daily Flows from Sophisticated (Low *ER*) Share Classes

Variable	Early crisis (1)	Peak crisis (2)	Lehman week (3)	Early crisis (4)	Peak crisis (5)	Lehman week (6)
$Low_{i,t-1}$	0.20	0.18	0.25	0.16	0.19	0.27
	(1.29)	(1.15)	(2.31)	(1.15)	(1.14)	(2.37)
$Low_{i,t-1} \times \%High_{i,t-2}$	−0.24	0.34	0.03	−0.41	0.14	−0.21
	(−0.60)	(0.52)	(0.07)	(−1.23)	(0.24)	(−0.74)
$High_{i,t-1}$	−0.04	−0.32	−0.16	−0.03	−0.27	−0.12
	(−0.39)	(−2.24)	(−1.76)	(−0.35)	(−1.82)	(−1.16)
$High_{i,t-1} \times \%High_{i,t-2}$	0.87	1.21	1.02	0.47	1.05	0.67
	(2.23)	(2.23)	(2.42)	(1.51)	(2.55)	(1.99)
$High_{i,t}$				−0.23	−0.10	−0.19
				(−2.43)	(−0.77)	(−1.56)
$High_{i,t} \times \%High_{i,t-1}$				1.59	1.53	1.64
				(4.76)	(2.45)	(3.11)
$\%Low_{i,t-1}$	0.78	−9.15	−4.58	0.19	−12.76	−7.14
	(0.42)	(−2.93)	(−2.96)	(0.11)	(−3.55)	(−4.55)
Average yield$_{i,t-1}$	−0.01	−0.71	−0.45	−0.14	−0.73	−0.43
	(−0.03)	(−0.53)	(−0.51)	(−0.39)	(−0.58)	(−0.59)
log total fund assets$_{i,t-1}$	−0.81	−0.15	−0.56	−0.83	−0.11	−0.55
	(−2.18)	(−0.17)	(−0.95)	(−2.15)	(−0.12)	(−0.98)
Avg institutional expense ratio$_{i,t-1}$	1.05	2.67	2.80	0.71	1.92	2.21
	(1.69)	(1.60)	(2.45)	(1.13)	(1.26)	(2.42)
Liquid asset share$_{i,t-1}$	0.40	1.90	1.29	0.45	1.82	1.38
	(0.90)	(1.55)	(1.70)	(1.04)	(1.49)	(2.05)
Fund business$_{i,t-1}$	−0.21	0.02	−0.10	−0.23	0.04	−0.05
	(−0.71)	(0.02)	(−0.17)	(−0.82)	(0.05)	(−0.09)
Observations	320	190	318	320	190	318
R^2	0.165	0.279	0.263	0.233	0.370	0.348

Notes: For each fund, we separate prime institutional share classes into two categories, based on their expense ratios. The first category, Low, consists of share classes which have expense ratios that are lower than the median expense ratio (across all institutional share classes within a given fund). All remaining share classes are included in the High category, including all retail share classes. The value of shares outstanding is then aggregated across all Low share classes and, separately, across all High share classes within a given fund. (Funds with a single share class are excluded from this analysis.) For each fund and date, we calculate the first difference in the log of aggregate value within each category (i.e., fraction flow), which we denote by $Low_{i,t}$ and $High_{i,t}$. The table presents the coefficients from panel regressions with $Low_{i,t}$ as the dependent variable on $Low_{i,t-1}$ and $High_{i,t-1}$, estimated for three different subperiods in 2008: 9/10–9/16 early crisis, 9/17–9/19 peak crisis, and 9/15–9/19 Lehman week, respectively. We multiply $Low_{i,t}$ and $High_{i,t}$ by 100 to express them in log percentage points. We also include interaction variables between, for example, $High_{i,t-1}$ and $\%High_{i,t-2}$, which is defined as two-day lagged fraction of total MMMF value within a MMMF represented by High (both institutional and all retail) *ER* share classes. Control variables, described in the notes to Table 3, have been divided by their (cross-sectional) standard deviations for ease of interpretation. All specifications also include unreported time dummies. Standard errors are clustered at the fund level. R^2 reports the overall R^2. t-statistics are reported in parentheses.

Table 6: Determinants of daily flows from sophisticated low-ER share classes.

6.7 Figure 4 and Table 7: Flow-distribution tails and quantile regressions

Read:

- Figure 4 plots the 10th, 50th, and 90th percentiles of daily flows for prime institutional and prime retail funds.
- The institutional distribution widens dramatically during Lehman week. The 10th percentile experiences very large outflows, while the median is much less extreme.
- Retail outflows are more muted and peak one day later than institutional outflows.

- Table 7 uses dynamic quantile regressions to simulate how fund characteristics shift cumulative Lehman-week flow quantiles.
- A one-standard-deviation increase in %Soph has a much larger effect on the 1st percentile than on the median, showing that sophisticated-investor concentration matters especially for tail run risk.
- Flow volatility and fund size also increase left-tail outflows, while average yield has a smaller but still negative effect in several quantiles.

Economic message: The paper's most policy-relevant object is not the average fund. It is the worst-outflow fund. Figure 4 and Table 7 show that run risk is a tail phenomenon, and investor composition is especially important in that tail.

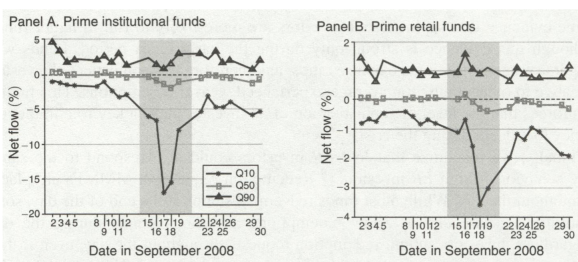

FIGURE 4. QUANTILES OF DAILY FLOW DISTRIBUTIONS BY CATEGORY DURING SEPTEMBER 2008

(a) Figure 4: Daily flow quantiles by category.

TABLE 7—MARGINAL EFFECTS OF FUND CHARACTERISTICS ON CUMULATIVE FLOW QUANTILES

		Cumulative flow quantile				
Variable	Value	1%	5%	10%	50%	90%
	$f(\bar{x})$	−52.02	−41.30	−35.81	−17.24	0.62
%Soph	$f(\bar{x}+\sigma_x)$	−62.30	−50.29	−44.18	−21.96	−0.47
	$f(\bar{x}-\sigma_x)$	−42.22	−32.87	−28.19	−12.93	2.59
	Difference	−20.08	−17.42	−15.99	−9.03	−3.06
	p-value	[0.003]	[0.002]	[0.001]	[0.001]	[0.170]
	p-value versus median	[0.020]	[0.019]	[0.018]	–	[0.046]
Average gross yield	$f(\bar{x}+\sigma_x)$	−55.04	−44.07	−38.51	−19.18	0.03
	$f(\bar{x}-\sigma_x)$	−48.92	−38.42	−33.04	−15.17	1.54
	Difference	−6.12	−5.65	−5.47	−4.02	−1.50
	p-value	[0.052]	[0.038]	[0.030]	[0.012]	[0.221]
	p-value versus median	[0.181]	[0.173]	[0.167]	–	[0.144]
log flow standard deviation	$f(\bar{x}+\sigma_x)$	−63.18	−50.45	−43.72	−19.36	10.86
	$f(\bar{x}-\sigma_x)$	−42.28	−33.14	−28.64	−14.42	−2.18
	Difference	−20.90	−17.31	−15.08	−4.94	13.04
	p-value	[0.000]	[0.000]	[0.000]	[0.013]	[0.014]
	p-value versus median	[0.000]	[0.000]	[0.000]	–	[0.000]
log fund total assets	$f(\bar{x}+\sigma_x)$	−56.93	−45.97	−40.31	−20.57	−0.32
	$f(\bar{x}-\sigma_x)$	−46.90	−36.32	−31.08	−13.49	2.74
	Difference	−10.02	−9.64	−9.23	−7.08	−3.06
	p-value	[0.042]	[0.019]	[0.011]	[0.001]	[0.163]
	p-value versus median	[0.257]	[0.235]	[0.221]	–	[0.090]

Notes: This table shows the impact of explanatory variables on cumulative flow distributions (as a percentage of initial assets) for prime institutional share classes (aggregated to the fund level) for the September 15–19 period. These estimates are obtained by simulating from an estimated dynamic quantile panel regression model for daily flows that is further described in the online Appendix. Columns report the 1st, 5th, 10th, 50th, and 90th quantiles of the cumulative flow distributions, respectively. We begin by fixing each of the explanatory variables at its average, assuming that the initial value of lagged flows equals the prime institutional category average. Then, we report the impact on the simulated flow distribution of adding and subtracting one standard deviation to each explanatory variable, as well as p-values for a test of whether the difference in the simulated quantiles is statistically significant, obtained by using the bootstrapped distribution of parameter estimates from our model, as well as the p-value of whether the marginal effect is significantly different at a given quantile, relative to the marginal effect at the median (using the bootstrapped distribution).

(b) Table 7: Marginal effects on cumulative flow quantiles.

7 Short summary

- The paper studies run-like behavior in money market mutual funds during September 2008.
- The main empirical innovation is daily share-class-level data, which allows comparisons between investor types inside the same fund.
- Low-expense-ratio institutional share classes proxy for sophisticated investors because they are associated with high minimum investment requirements.
- Sophisticated investors redeem more aggressively during Lehman week.
- Funds with a larger fraction of sophisticated investors experience larger outflows.
- The outflow gap between sophisticated and less sophisticated investors is larger in funds with more sophisticated investors.

- Daily regressions show feedback between investor types: sophisticated investors react to less sophisticated investor flows when those investors are important in the fund.
- Ultra-short bond fund evidence suggests that fixed NAVs amplified first-mover incentives in MMMFs.
- Quantile regressions show that investor composition matters especially in the left tail of fund flows.
- The paper's broader message is that shadow-bank runs depend jointly on asset quality, liability structure, and the composition of investors.

8 Conclusion

8.1 Core conclusion

Schmidt, Timmermann, and Wermers show that the September 2008 money market fund run cannot be understood only as a reaction to bad assets. Investor composition played a central role. Large, sophisticated institutional investors were more likely to redeem, and their incentives were stronger when other sophisticated investors were also present in the same fund. The evidence is consistent with strategic complementarities: each investor's incentive to run depends on the expected behavior of other investors.

8.2 Economic intuition

The intuition is that a prime money fund is a pooled vehicle with cash-like liabilities and imperfectly liquid assets. If some investors redeem early, the fund may be forced to use its liquid assets or sell securities under stress. Remaining investors then face a worse portfolio and greater risk of loss. Sophisticated investors understand this externality and therefore have a stronger incentive to move quickly. When a fund contains many sophisticated investors, each one expects the others to move quickly as well, so the run equilibrium becomes more likely.

8.3 Incremental contribution revisited

The paper advances the empirical literature on financial runs in three ways. First, it uses unusually granular daily share-class data to observe heterogeneous investors inside the same portfolio. Second, it turns a theoretical model of strategic complementarities into concrete empirical predictions about flows. Third, it shows that run risk is concentrated in the tail and is amplified by the fixed-NAV structure of money market funds.

8.4 Implications for regulation and investors

The paper suggests that regulation should focus not only on asset holdings but also on investor composition and payoff structure. Liquidity requirements address asset-side fragility, but liability-side fragility can remain if investors are sophisticated, concentrated, and able to redeem quickly. Floating NAVs may reduce first-mover advantages, but separating institutional and retail investors can also concentrate sophisticated investors together, potentially strengthening strategic complementarities.

8.5 Limitations

The study is observational, so investor composition is not randomly assigned. Expense ratios are a proxy for sophistication rather than a direct measure of investor information. Some unobserved portfolio risk may remain despite fund fixed effects and controls. The ultra-short bond fund comparison is informative but imperfect because ultra-short funds differ from MMMFs in assets, investors, and accounting. Finally, the crisis setting is extreme, so the exact magnitudes should not be mechanically extrapolated to normal periods.

8.6 Takeaway

The enduring takeaway is that runs are social and strategic, not purely fundamental. In cash-like investment pools, the question is not only "What assets does the fund hold?" but also "Who else is invested, how quickly will they react, and what happens to me if they leave first?" This paper makes that mechanism visible using the share-class structure of money market mutual funds.